THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. When a model fails in some situation, the scientist: 1

(a) discards all science
(b) ignores the failure
(c) blames observers
(d) defines its limits and develops a better model

2. A scientist measures 12.5 g, 12.7 g and 12.6 g for the same mass. The mean is: 1

(a) 12.7 g **(b)** 12.5 g **(c)** 12.6 g **(d)** 37.8 g

3. If 1 litre is 1000 cubic cm, then 2.5 litres equals: 1

(a) 2500 cubic cm **(b)** 250 cubic cm **(c)** 25000 cubic cm **(d)** 25 cubic cm

4. Which of the following is NOT a testable hypothesis? 1

(a) 'Beauty is the best colour of a flower.' **(b)** 'Heated gas expands.'
(c) 'More light increases plant growth.' **(d)** 'Salt dissolves faster in hot water.'

5. Percentage error when the measured value is 49 g and the true is 50 g: 1

(a) 2 per cent **(b)** 1 per cent **(c)** 0.2 per cent **(d)** 5 per cent

6. A scientist finds that her data do not fit the model. The best step is to: 1

(a) stop work **(b)** publish only matching data
(c) check the data and refine the model **(d)** throw away the data

7. Which one is a quantitative observation? 1

(a) The liquid has a mass of 30 g. **(b)** The liquid is clear.
(c) The liquid is blue. **(d)** The liquid smells sweet.

8. An estimate of 3 x 10^8 m/s for the speed of light written in km/s is: 1

(a) 3 x 10^2 **(b)** 3 x 10^5 **(c)** 3 x 10^8 **(d)** 3 x 10^11

9. A reading of a stopwatch 12.43 s has how many significant figures? 1

(a) 3 **(b)** 4 **(c)** 5 **(d)** 2

10. If a scale shows 0.2 cm at no load (zero error), the true length for a reading of 5.4 cm is: 1

(a) 5.2 cm **(b)** 5.4 cm **(c)** 5.6 cm **(d)** 0.2 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the percentage error if the measured length is 9.8 cm and the true length is 10 cm. 2

12. Why are the symbols for physical quantities printed in italics and units in plain form? 2

13. The length of a wire is 150 cm. Express it in 2
(a) m
(b) mm.

14. Define least count. What is the least count of a metre rule with millimetre markings? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain, with an example, how a hypothesis may be modified after an experiment. 3

16. Explain the sources of error in a stopwatch measurement of the time to cycle from school to home and how to reduce them. 3

17. Give three examples of laws of science and the phenomenon they describe. 3

OR

State three differences between precision and accuracy with an example of a faulty balance. 3

18. The claim 'tomatoes are red because of ripening' is tested by ripening tomatoes in dark and light. Identify the variables and expected result. 3

OR

Explain why different scientists in different countries need standard symbols and notations. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Discuss the development of the atomic model from Dalton to Bohr as an example of scientific models evolving with evidence. 5

OR

Explain the process of scientific communication and why honesty, humility and cooperation matter, with two examples. 5

20. Compare accuracy, precision, least count and zero error with examples, and describe how to correct a reading. 5

OR

A student claims that a new fertiliser doubles yield after growing it on one plot. Evaluate the claim and suggest a proper test. 5

END OF QUESTION PAPER — Total: 40 Marks

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